Roll No.....

ODD MID SEMESTER EXAMINATION SEPTEMBER - 2025

Name of the Program: **M. TECH** | *Semester:* **I**

Name of the Course: **Electric Vehicle Technology** | *Course Code:* **MEV 105**

Time: 90 Minutes | Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

Q1	(10 marks)	CO 1
a)	Explain the working of electric vehicle (EV) with a block diagram. What are the essential components of an EV? Give a comparative analysis between EV and internal combustion (IC) engine vehicle.	
	OR	
b)	What are the classifications of EV? Also state the advantages and disadvantages of EV.	
Q2	(10 marks)	CO 1
a)	Define the regenerative breaking phenomenon in plug-in hybrid electric vehicle (PHEV). Explain the working and the main components of PHEV with a suitable block diagram.	
	OR	
b)	What is the concept of hybrid electric vehicle (HEV)? How HEVs are different from PHEVs? What is the importance of the Propulsion System in HEVs? Also state the main components of the propulsion system.	
Q3	(10 marks)	CO 1
a)	How series, parallel, and series-parallel HEVs are different from each other, explain with the help of the suitable block diagram.	
	OR	
b)	Explain the working of Fuel Cell Electric Vehicle (FCEV) with a suitable block diagram. What are the advantages and disadvantages of FCEVs?	
Q4	(10 marks)	CO 2
a)	State Newton's Second Law of motion with the appropriate equation. Define and explain different type of forces or resistances exist during the motion of the vehicle.	
	OR	
b)	For a vehicle with $M_v = 1500$ kg, $A*C_d = 0.7$ m^2, $\mu = 0.012$, vehicle speed (v) = 120 km/h, and an acceleration (a) = 0.027g. Calculate the traction torque required at the wheels and the corresponding rotational speed level (tires 195/65/15T). Also calculate the road slope that is equivalent to that acceleration.	
Q5	(10 marks)	CO 2
a)	Define the following terms with suitable formula related to the drive train of an EV: (i) Torque transmitted from the power plant to the drive wheels (ii) Tractive effort on the driven wheels (iii) Rotating speed of the driven wheel (iv) Translational speed of the wheel (vehicle speed)	
	OR	
b)	Give classifications of motors used in EVs. Also compare these motors based on their advantages and disadvantages.	